Post-Quantum Cryptography Migration

Part 4 - Cryptographic Foundations: Crypto Libraries & Providers

Compiled by the Spaced Research Ledger

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Almost no application implements cryptography itself. It calls a library. A few dozen libraries therefore decide when the world's software actually becomes quantum-resistant, whatever the standards say.

This report covers that layer in six sections. Section 1 covers OpenSSL and its forks, Section 2 the reference implementations everything borrows from, and Section 3 the language runtimes with their own stacks. Section 4 covers the embedded and alternative libraries, Section 5 the plumbing that connects libraries to hardware, and Section 6 the first attacks on the new implementations.

The center of gravity has shifted from add-ons to defaults. OpenSSL 3.5 ships the three NIST algorithms natively and prefers hybrid post-quantum key exchange out of the box. Go, Java, and .NET now carry the algorithms in their standard libraries. The pioneering scaffolding is being retired on schedule, with the PQClean reference collection archived and wolfSSL dropping its last add-on dependency.

The other shift is that attackers have arrived. Nobody has broken the standardized algorithms themselves. But 2026 brought the first practical attacks on implementations, including a demonstration recovering ML-DSA keys from buggy software in under a second. The new battleground is code quality, not mathematics.

1. OpenSSL and Forks

OpenSSL is the default cryptography library of the internet's server side, and its forks guard specific empires: BoringSSL inside Google and Chrome, AWS-LC and s2n-tls inside Amazon, LibreSSL inside OpenBSD. When these projects change a default, millions of servers change behavior on their next update. This section covers how each has taken up the new algorithms.

Recent Developments

The forks are moving from key exchange into signatures and certification. AWS-LC holds multiple active FIPS 140-3 certificates with ML-KEM inside validated modules [1]. BoringSSL now supports ML-DSA signatures at all three parameter sets [2]. Amazon's s2n-tls supports ML-DSA certificates for TLS 1.3 authentication and includes them in its post-quantum and CNSA policies [3], via a specific dated policy version [3].

Current State

OpenSSL itself crossed the line in April 2025. Version 3.5, a long-term-stable release, added native ML-KEM, ML-DSA, and SLH-DSA, and changed the default TLS group list to prefer hybrid post-quantum key exchange [4]. Support runs through April 2030 [5]. Distributions inherit it directly, with RHEL shipping the algorithms in its default provider [6].

OpenSSL 3.6 added verification for the LMS hash-based scheme [7]. For what 3.5 does not natively offer in TLS, such as SLH-DSA, the third-party oqs-provider remains the bridge [8].

Amazon's stack is built for FIPS. AWS-LC implements ML-KEM and ML-DSA at every parameter set [9], imported the verified mlkem-native implementation into its FIPS module [10], and added ML-DSA as a TLS 1.3 signature scheme [10]. Its FIPS 3.0 module was the first open-source cryptographic module to enter NIST's validation pipeline with ML-KEM in scope [11]. One gap: the higher-level Rust binding had not fully stabilized ML-DSA as of February 2026, leaving downstream users on a separate crate [12].

Above the library, s2n-tls implements the hybrid handshakes used across AWS service endpoints [13], with clients able to prefer hybrid while keeping classical fallback [13]. The hybrid suites are supported only on Linux [13]. A NIST presentation confirms the division of labor: AWS-LC provides the algorithms, s2n-tls the handshake, integrated into AWS's load balancers [14].

Google's fork moved earliest. BoringSSL has carried the X25519 plus ML-KEM hybrid since 2023, under its pre-standard name at first [15]. Chrome shipped the draft group by default in April 2024 and switched to the standardized codepoint that November [16]. Google plans to retire the enterprise escape hatch by Chrome 147, making hybrid key exchange effectively mandatory in that ecosystem [17].

LibreSSL followed its own path, adding an ML-KEM API in late 2025 and the hybrid TLS keyshare in April 2026 [18]. All of this builds on NIST's August 2024 finalization, which replaced the old Kyber and Dilithium names [19].

Unresolved Conflicts

Two dating disputes stand. One source records BoringSSL adding ML-DSA support through specific commits in October 2025 [20], while another dates the addition to around April 2026 without superseding the earlier evidence [21]. The commit history is public, so a direct check settles it. Separately, a comparison piece frames AWS-LC and BoringSSL as diverging on default-on behavior [22]. It rests on an AWS-LC version claim that could not be verified against the project's own release notes [22].

2. PQ Reference Implementations

Before algorithms reach products, they live in reference implementations: carefully written, often formally verified code that libraries import rather than reimplement. A handful of projects fill this role, led by liboqs and the PQ Code Package. This section covers a generational handover, as the research-era scaffolding retires in favor of verified, production-grade successors.

Recent Developments

The era's most important cleanup happened on schedule. PQClean, the collection that made NIST reference code usable since 2019, was archived read-only on August 4, 2026 [23]. Downstream Rust crates wrapping its code are now marked unmaintained in the security advisory database, since no further patches will land upstream [24].

liboqs consolidated in the same release cycle. Version 0.16.0 removed SPHINCS+, renamed its FrodoKEM variants, and fixed an out-of-bounds read and an uninitialized-pointer bug [25]. It moved HQC to the algorithm's official upstream and enabled it by default [26], made mldsa-native the default ML-DSA [26], and updated mlkem-native to version 1.2.0 [26]. That mlkem-native release added a PowerPC assembly backend, broadened portability, hardened RISC-V timing, and extended its formal proofs [27]. A matching oqs-provider release is pending in an open issue [28].

Current State

PQClean's retirement was orderly. Its README now redirects users to the PQ Code Package projects and liboqs [29]. The maintainers proposed retirement in January 2026 and confirmed the plan on the NIST mailing list [30]. They noted the project could never keep pace with the signature on-ramp or the finalized standards, with a lessons-learned paper on ePrint [31].

Researchers already cite it at pinned commits as a frozen artifact [32]. liboqs is discussing the operational impact, since it relied on PQClean for several algorithms [33].

The successors are verification-first. mlkem-native's C code is proved memory-safe and type-safe, and its ARM and x86 assembly functionally correct and constant-time [34]. It became liboqs's default ML-KEM in three variants [25], first shipping stable in liboqs 0.14.0 [35]. Its own soundness notes are candid that verification is never absolute, with named coverage gaps [36].

A version 2.0.0 introduced breaking API changes, with a long-term-support branch until February 2027 [37]. The signature sibling, mldsa-native, remains a production-ready alpha [38]. Apple evaluated both as prior art for its own verification effort and built a custom approach instead [39].

liboqs itself keeps shedding pre-standard history. Version 0.15.0 was the last with SPHINCS+ and removed the pre-standardization Dilithium [40].

An HQC timing vulnerability was fixed by disabling the offending compiler optimizations, with HQC off by default at the time [25]. The oqs-provider companion has kept pace. Recent releases added new signature candidates and Brainpool hybrids, and disabled overlapping algorithms when OpenSSL 3.5 is present [41]. Version 0.11.0 shipped at the end of 2025 [42], and the development branch re-instated HQC [43].

3. Language-Native Crypto Stacks

Most developers never call a C library directly. They use whatever their language runtime provides. So the migration only reaches everyday software when Go, Java, .NET, and Rust ship the algorithms in their standard libraries. This section covers that arrival, which is now essentially complete for key exchange and well underway for signatures.

Recent Developments

Go completed its set in August. Go 1.27 adds a standard-library ML-DSA package at all three parameter sets, with ML-DSA keys and signatures in certificates and TLS [44]. The same release adds the ML-KEM-1024 key exchange [44].

Java committed to its long-term releases. JDK 27 introduces hybrid post-quantum TLS key exchange, completing the core capabilities begun in JDK 24 [45]. Oracle plans backports of the algorithms and the hybrid exchange to JDK 25 in October 2026, with older long-term releases following in 2027 [45]. Bouncy Castle's 1.85 release added hybrid certificates, post-quantum keys and signatures in X.509, and standards-based migration mechanisms [46]. In .NET, a breaking rename aligned the new algorithm APIs with the platform's conventions [47].

Current State

Go made post-quantum the path of least resistance. Version 1.24 enabled hybrid X25519 plus ML-KEM by default whenever an application does not set preferences [48]. The current default order runs three hybrid groups ahead of every classical curve [49]. Version 1.26 added standard hybrid public-key encryption, including the X-Wing construction [50].

The default propagates transitively: Kubernetes gained post-quantum TLS simply by building with Go 1.24 [51]. One FIPS-mode inconsistency, a stale entry making the default hybrid unreachable under the strictest setting, is filed with a recommended fix [52].

Java's platform support arrived in JDK 24 as native engine classes for ML-KEM and ML-DSA [53]. JDK 26 extended ML-DSA to JAR signing and added hybrid public-key encryption APIs [54]. The key tool can generate ML-KEM certificates, though a KEM key still needs a separate signature algorithm to sign its certificate [55]. A private-key encoding change in recent patch releases breaks readability by older JDKs unless a property is set back [56].

Bouncy Castle layered in earlier, with ML-KEM in 1.78, ML-DSA in 1.79, and composite signatures needing 1.80 [22]. TLS support for ML-KEM suites came in 1.81 [57]. A FIPS-track library is in early access [58], and a specification page separates standardized algorithms from experimental candidates [59].

.NET 10 shipped all three NIST standards plus composite ML-DSA in its standard cryptography namespace, with no extra packages [60]. The APIs need Windows with CNG support or Linux with OpenSSL 3.5, and explicitly exclude macOS [60]. The underlying Windows path came through SymCrypt and CNG [61], generally available since November 2025 [62], with certificate issuance following [61]. Some interoperability helpers remain experimental behind a warning [63].

Rust's mainstream TLS stack folded the work in. Since rustls 0.23.22, ML-KEM hybrid key exchange lives in the core crate behind a feature flag [64], superseding a separate crate still labeled experimental [65]. AWS documents enabling it as a one-line feature flag for its SDK users [66].

4. Embedded & Alternative Libraries

Outside the OpenSSL orbit sits a second ecosystem: the small, portable libraries that run on routers, industrial controllers, and anything with kilobytes rather than gigabytes. Here one vendor has sprinted ahead while some of the most widely deployed embedded libraries have barely started. This section covers both extremes.

Recent Developments

wolfSSL turned August into a product barrage. One announcement covered seven simultaneous developments, including a post-quantum FIPS module, kernel integration, quantum-ready WireGuard and Tailscale, and a satellite cryptographic module under NIST test [67]. The library finished removing liboqs entirely, going fully native for every algorithm it ships, FrodoKEM included [68]. Its implementations gained NIST algorithm validation [69].

Around those, the same vendor shipped an OpenSSL provider bridge [70] and a release renaming Dilithium to ML-DSA with extended certificate support [71]. It also added post-quantum support in its PKCS#11 token [72], its Java providers [73], and curl integration [74].

Botan stays current more quietly, with its latest stable release continuing broad coverage across the NIST algorithms, hash-based schemes, and hybrid TLS [75].

Current State

wolfSSL's standing position is early-adopter for embedded and government work. Its wolfCrypt engine has full TLS 1.3 support for ML-KEM and ML-DSA [76]. Its CNSA-compliant set spans the three standards plus LMS and XMSS, with assembly optimizations across architectures [77]. It implements the new TPM specification's post-quantum commands [78], ships a firmware TPM validated against NIST test vectors [79], and has post-quantum algorithms on its Rust API roadmap [80].

Botan earned an unusual credential: Germany's federal cybersecurity agency confirmed a reviewed version contains every post-quantum scheme its national technical guideline recommends [81]. The library added the three NIST standards in release 3.6.0 and enabled hybrid key exchange by default in 3.7.0 [82], with seed-based key storage per the FIPS specifications [83].

The middle of the field is uneven. GnuTLS added experimental hybrid key exchange [84], then experimental ML-DSA with an alternative backend [85], preferred for its seed-based key generation [86]. RHEL gates its GnuTLS post-quantum support behind a subpolicy [87]. Its current release added a key-derivation refinement [88], though the version claim itself comes from a secondary source [89].

Mozilla's NSS vendored a formally verified Rust ML-KEM early [90] and made the hybrid group default in late 2025 while adding ML-DSA interfaces [91]. Firefox inherits it [91], with the current version again cited from a secondary source [92].

The laggards matter because of their reach. libsodium shipped nothing post-quantum through mid-2025 [93], then delivered its first primitive in April 2026: ML-KEM-768 plus the X-Wing hybrid, which its notes recommend for most applications [94]. Mbed TLS, the default on a huge share of embedded devices, still ships no finalized post-quantum algorithm [93]. Its roadmap schedules initial ML-DSA for mid-2026 and puts ML-KEM, the algorithm TLS needs most, in an unscheduled future bucket [95]. Community threads show the frustration persisting [96].

5. PKCS#11 and Provider Plumbing

Between a library and a hardware security module sits plumbing: the PKCS#11 token interface, OpenSSL's provider architecture, Windows key-storage providers, Java's pass-through layer. Algorithms only reach protected hardware when this plumbing names them. This section covers the standard that finally did, and the vendors racing to implement it.

Recent Developments

The standard arrived. OASIS ratified PKCS#11 version 3.2 in July 2026, adding post-quantum mechanism support and the conformance profiles for interoperable implementations [97]. The specification defines concrete mechanisms and key types for ML-KEM, ML-DSA, and SLH-DSA, plus new encapsulation functions [98], with the full text published in June [99]. IBM announced its openCryptoki project already implements it, and Entrust welcomed the ratification [97].

Vendors updated in step. Thales published ML-KEM and ML-DSA programming guides using proprietary extensions pending full standardization [100]. Securosys [101] and Fortanix [102] confirmed post-quantum support through their PKCS#11 providers. wolfSSL added the v3.2 algorithms to its token [103], Mastercard's open-source tools now generate post-quantum keys and certificates by default [104], and oqs-provider development continues adding hybrid codepoints [105].

Current State

Hardware tokens got there first. Thales Luna firmware 7.9.0 integrated ML-KEM and ML-DSA natively through standard PKCS#11 [106]. Extensions added quantum-resistant key wrapping and attestation [107], multi-module LMS state management [108], and a custom-silicon Luna 8 line for throughput [109]. The open-source SoftHSMv2 grew build flags for both algorithms [110], answering a request opened in mid-2025 [111].

An independent software HSM project targets PKCS#11 users and DNSSEC operators directly [112]. Adjacent to the token world, AWS KMS signs with ML-DSA at all parameter sets [113].

The standardization tail matters for compatibility. LMS entered PKCS#11 in v3.1, with the lattice algorithms slated for v3.2 [114]. The v3.2 drafts progressed through 2025 [115], and current integration documentation notes the ratified version omits the pre-hashed ML-DSA variant [116].

On the software-provider side, OpenSSL 3.5's native support [117] coexists with oqs-provider's broader coverage [118]. That provider plugs in through OpenSSL's provider API, not PKCS#11 [119]. On Linux, p11-kit aggregates multiple PKCS#11 modules behind one proxy [120].

Windows and Java complete the map. Microsoft exposed the algorithms through CNG from Insider builds [121] to general availability [62], with certificate services issuing ML-DSA certificates from a newly built hierarchy [122].

Production hardware-backed keys depend on each HSM vendor's own key-storage provider [123]. The hybrid TLS groups shipped in July 2026, disabled by default [124], and architecture write-ups frame provider selection as the central practical decision [125]. Java's SunPKCS11 is a pure conduit, exposing whatever the underlying vendor module advertises [126]. The JDK's own providers carry ML-KEM and ML-DSA natively since JDK 24 [127], with measurable performance gains in JDK 25 [128].

6. PQ Implementation Vulns & Cryptanalysis

A new algorithm can be mathematically sound and still fall to a bug in the code that runs it. Two years into deployment, that is exactly the pattern: the standards stand unbroken while researchers pick apart implementations. This section covers what has actually been attacked, what was fixed, and what remains theoretical.

Recent Developments

The implementation attacks got sharp in 2026. Daniel J. Bernstein published working attacks that recover ML-DSA secret keys from two classes of implementation bugs, in software including OpenSSL and mldsa-native [129]. The forgeries take under a second on one core [129].

A cache-timing attack on the optimized HQC implementation, exploiting compiler-induced branches, achieved full key recovery and was presented at Crypto 2026 [130]. OpenSSL's August 2026 advisory includes a moderate heap overflow in CMS key unwrapping reachable through ML-KEM decapsulation [131]. liboqs 0.16.0 shipped its own memory-safety fixes [132].

The baselines stay clean where it matters most. No published cryptanalytic break of the ratified ML-KEM, ML-DSA, or SLH-DSA parameter sets has been found [133]. No CVE specific to BoringSSL's post-quantum code exists [134], and none specific to Bouncy Castle's [135].

Current State

The pattern established itself early: timing bugs, caught and fixed. KyberSlash exploited non-constant-time arithmetic in Kyber decapsulation and was patched in the reference code by December 2023 [133]. The Clangover finding showed compilers can reintroduce secret-dependent branches on their own, with liboqs's HQC code leaking keys only when built with Clang at higher optimization [136]. liboqs fixed it in 0.14.0 [137], noting the bug only hit users who explicitly enabled HQC [138].

An earlier HQC decapsulation bug had already been fixed [137]. The same 0.14.0 release brought the formally verified mlkem-native in as the default, which is the structural answer to this bug class [138].

HQC itself carries the one open design-level question. An advisory documents a theoretical flaw in its implicit-rejection mechanism, with no concrete attack known and no patch available [139]. The algorithm stays disabled by default pending an updated specification [139]. HQC is NIST's backup KEM, not one of the three ratified standards [140].

The big vendors' incidents have been adjacent to, not inside, the post-quantum code. Three AWS-LC vulnerabilities fixed in early 2026 involved certificate-chain validation bypasses and a timing leak [141], the worst skipping validation of all but the final signer [142]. Amazon reported no service impact [141].

AWS-LC's post-quantum footprint itself remains broad and on the FIPS validation track [143][144]. BoringSSL's only recorded CVE predates its post-quantum work entirely [145], and its ML-KEM rollout through Chrome completed without incident [133].

Bouncy Castle's post-quantum line has been feature work [146][147], with one classical-adjacent fix: a non-constant-time comparison in FrodoKEM [148]. OpenSSL's one recorded KEM advisory concerns classical RSA-KEM, not ML-KEM [149], while its 3.5 defaults [150] ride a support window to 2030 [151]. One ecosystem-wide argument continues: OpenSSL, Bouncy Castle, NSS, and Red Hat have jointly asked the IETF to reconsider the seed-only private-key format over FIPS-certification concerns [152].

About This Report

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Sources

1. Cryptographic Module Validation Program | CSRC - https://csrc.nist.gov/projects/cryptographic-module-validation-program/certificate/4631 - as of 2026-08-25
2. PQC support · Cloudflare SSL/TLS docs - https://developers.cloudflare.com/ssl/post-quantum-cryptography/pqc-support/ - as of 2026-06-24
3. Post Quantum Support - https://aws.github.io/s2n-tls/usage-guide/ch16-post-quantum.html - as of 2026-06-15
4. postquantum.com — openssl 3 5 pqc default - https://postquantum.com/security-pqc/openssl-3-5-pqc-default/ - as of 2025-04-08
5. openssl-corporation.org — post quantum - https://openssl-corporation.org/post-quantum.html
6. access.redhat.com — 7119430 - https://access.redhat.com/articles/7119430
7. OpenSSL 3.6 Series Release Notes | OpenSSL Library - https://www.openssl.org/news/openssl-3.6-notes.html - as of 2025-10-01
8. github.com — oqs provider - https://github.com/open-quantum-safe/oqs-provider
9. github.com — PQREADME - https://github.com/aws/aws-lc/blob/main/crypto/fipsmodule/PQREADME.md
10. github.com — releases - https://github.com/aws/aws-lc/releases
11. aws.amazon.com — aws lc fips 3 0 first cryptographic library to include ml kem in fips 140 3 validation - https://aws.amazon.com/blogs/security/aws-lc-fips-3-0-first-cryptographic-library-to-include-ml-kem-in-fips-140-3-validation
12. github.com — 17 - https://github.com/LatticeArc/latticearc/issues/17 - as of 2026-02-08
13. docs.aws.amazon.com — pqtls - https://docs.aws.amazon.com/kms/latest/developerguide/pqtls.html

14. csrc.nist.gov — buillding pw cloud services.pdf - https://csrc.nist.gov/csrc/media/Presentations/2025/building-post-quantum-cloud-services/images-media/buillding-pw-cloud-services.pdf
15. quantumsecuritydefence.com — fips 203 205 implementation c rust libraries - https://quantumsecuritydefence.com/insights/fips-203-205-implementation-c-rust-libraries/
16. netguardia.com — hybrid key exchange today why x25519 ml kem is the interim default - https://netguardia.com/privacy/encryption/hybrid-key-exchange-today-why-x25519-ml-kem-is-the-interim-default/
17. pbxscience.com — chromes quantum lock how ml kem changed the web - https://pbxscience.com/chromes-quantum-lock-how-ml-kem-changed-the-web/ - as of 2026-04-01
18. ianix.com — pqcrypto deployment - https://ianix.com/pqcrypto/pqcrypto-deployment.html - as of 2026-04-18
19. vucense.com — ml kem fips 203 implementation guide 2026 - https://vucense.com/privacy-sovereignty/post-quantum-cryptography/ml-kem-fips-203-implementation-guide-2026/
20. BoringSSL upgrade for Post-Quantum Cryptography support - https://github.com/envoyproxy/envoy/issues/42893 - as of 2026-01-07
21. Cryptography | Noise - https://noise.getoto.net/tag/cryptography - as of 2026-04-01
22. evertrust.io — hybrid post quantum certificates - https://evertrust.io/blog/hybrid-post-quantum-certificates/ - as of 2026-06-03
23. GitHub - PQClean/PQClean - https://github.com/pqclean/PQClean - as of 2026-08-04
24. rustsec.org — RUSTSEC 2026 0163 - https://rustsec.org/advisories/RUSTSEC-2026-0163 - as of 2026-06-04
25. github.com — releases - https://github.com/open-quantum-safe/liboqs/releases - as of 2026-07-09
26. liboqs/RELEASE.md - https://github.com/open-quantum-safe/liboqs/blob/main/RELEASE.md - as of 2026-07-09
27. mlkem-native/RELEASE.md - https://github.com/pq-code-package/mlkem-native/blob/main/RELEASE.md - as of 2026-08-10
28. Release new oqs-provider version · Issue #797 - https://github.com/open-quantum-safe/oqs-provider/issues/797 - as of 2026-07-09
29. github.com — PQClean - https://github.com/pqclean/PQClean
30. github.com — Z30 l8mTNRU - https://github.com/PQClean/PQClean/issues/604, corroborated by https://groups.google.com/a/list.nist.gov/g/pqc-forum/c/Z30-l8mTNRU - as of 2026-01-08
31. groups.google.com — Z30 l8mTNRU - https://groups.google.com/a/list.nist.gov/g/pqc-forum/c/Z30-l8mTNRU
32. arxiv.org — 2603.19340v1 - https://arxiv.org/html/2603.19340v1 - as of 2026-03-19
33. github.com — 2341 - https://github.com/open-quantum-safe/liboqs/issues/2341 - as of 2026-01-08
34. github.com — mlkem native - https://github.com/pq-code-package/mlkem-native
35. pqca.org — pqca announces release of liboqs version 0 14 0 from open quantum safe project - https://pqca.org/blog/2025/pqca-announces-release-of-liboqs-version-0-14-0-from-open-quantum-safe-project/ - as of 2025-07-10

36. github.com — SOUNDNESS - https://github.com/pq-code-package/mlkem-native/blob/main/SOUNDNESS.md
37. mlkem-native/RELEASE.md at main · pq-code-package/mlkem-native · GitHub - https://github.com/pq-code-package/mlkem-native/blob/main/RELEASE.md
38. pqca.org — issues - https://pqca.org/blog/2025/pqca-announces-alpha-release-of-mldsa-native/, corroborated by https://github.com/pq-code-package/mldsa-native/issues - as of 2025-11-25
39. github.com — formal verification for apple corecrypto - https://github.com/apple/corecrypto/blob/2026-05/corecrypto_verify/technical_overview/formal-verification-for-apple-corecrypto.md
40. github.com — RELEASE - https://github.com/open-quantum-safe/liboqs/blob/main/RELEASE.md - as of 2025-11-14
41. github.com — RELEASE - https://github.com/open-quantum-safe/oqs-provider/blob/main/RELEASE.md
42. Releases · open-quantum-safe/oqs-provider - https://github.com/open-quantum-safe/oqs-provider/releases - as of 2025-12-24
43. oqs-provider/RELEASE.md at main · open-quantum-safe/oqs-provider · GitHub - https://github.com/open-quantum-safe/oqs-provider/blob/main/RELEASE.md
44. Go 1.27 Release Notes - https://go.dev/doc/go1.27 - as of 2026-08-22
45. Post-Quantum Cryptography in Long-Term Support JDK Releases - https://blogs.oracle.com/java/post-quantum-cryptography-in-long-term-support-jdk-releases - as of 2026-08-06
46. New Release: Bouncy Castle Java 1.85 - https://www.bouncycastle.org/resources/new-release-bouncy-castle-java-1-85/ - as of 2026-07-28
47. Breaking change - MLDsa and SlhDsa 'SecretKey' members renamed - https://learn.microsoft.com/en-us/dotnet/core/compatibility/cryptography/10.0/mldsa-slhdsa-secretkey-to-privatekey - as of 2026-07-31
48. pkg.go.dev — tls - https://pkg.go.dev/crypto/tls
49. golang.bg — defaults.go - https://golang.bg/src/crypto/tls/defaults.go
50. medium.com — go 1 26 new crypto packages security built for the post quantum era bfff794fd09a - https://medium.com/techtrends-digest/go-1-26-new-crypto-packages-security-built-for-the-post-quantum-era-bfff794fd09a - as of 2026-02-19
51. kubernetes.io — pqc in k8s - https://kubernetes.io/blog/2025/07/18/pqc-in-k8s/
52. github.com — 78178 - https://github.com/golang/go/issues/78178 - as of 2026-03-16
53. baeldung.com — jdk security enhancements - https://www.baeldung.com/jdk-security-enhancements - as of 2026-06-05
54. seanjmullan.org — jdk26 - https://seanjmullan.org/blog/2026/03/16/jdk26 - as of 2026-03-16
55. neatguycoding.com — 2026 05 18 javaone 2026 java and post quantum cryptography - https://neatguycoding.com/posts/2026-05-18-javaone-2026-java-and-post-quantum-cryptography/ - as of 2026-05-18
56. oracle.com — 26 0 2 relnotes - https://www.oracle.com/java/technologies/javase/26-0-2-relnotes.html

57. Bouncy Castle Releases Java 1.81 and C# .NET 2.6.1 - https://www.bouncycastle.org/resources/bouncy-castle-releases-java-1-81-and-c-net-2-6-1/ - as of 2025-06-09
58. bouncycastle.org — roadmap - https://www.bouncycastle.org/download/bouncy-castle-java-fips/roadmap/
59. bouncycastle.org — specification interoperability - https://www.bouncycastle.org/documentation/specification_interoperability/ - as of 2026-03-10
60. duendesoftware.com — 20260514 post quantum cryptography in dotnet 10 - https://duendesoftware.com/blog/20260514-post-quantum-cryptography-in-dotnet-10 - as of 2026-05-14
61. kryptus.com — appnote adcs ml dsa knet en.pdf - https://kryptus.com/wp-content/uploads/2026/06/appnote-adcs-ml-dsa-knet-en.pdf
62. techcommunity.microsoft.com — 4469093 - https://techcommunity.microsoft.com/blog/microsoft-security-blog/post-quantum-cryptography-apis-now-generally-available-on-microsoft-platforms/4469093 - as of 2025-11-18
63. medium.com — post quantum cryptography in net 10 4da2779dcd2a - https://medium.com/@anderson.buenogod/post-quantum-cryptography-in-net-10-4da2779dcd2a - as of 2026-01-12
64. docs.rs — rustls post quantum - https://docs.rs/rustls-post-quantum/latest/rustls_post_quantum/
65. lib.rs — rustls post quantum - https://lib.rs/crates/rustls-post-quantum
66. aws.amazon.com — ml kem post quantum tls now supported in aws kms acm and secrets manager - https://aws.amazon.com/blogs/security/ml-kem-post-quantum-tls-now-supported-in-aws-kms-acm-and-secrets-manager/ - as of 2025-04-07
67. prweb.com — wolfssl advances post quantum encryption and fips 140 3 across embedded and enterprise systems to facilitate cnsa 2 0 compliance 302840890 - https://www.prweb.com/releases/wolfssl-advances-post-quantum-encryption-and-fips-140-3-across-embedded-and-enterprise-systems-to-facilitate-cnsa-2-0-compliance-302840890.html - as of 2026-08-04
68. wolfCrypt Retires liboqs as Native Post-Quantum Support Expands - https://www.wolfssl.com/wolfcrypt-retires-liboqs-as-native-post-quantum-support-expands/ - as of 2026-08-19
69. Post-Quantum - wolfSSL - https://www.wolfssl.com/category/post-quantum/ - as of 2026-08-13
70. wolfProvider Post-Quantum Cryptography: ML-KEM and ML-DSA - wolfSSL - https://www.wolfssl.com/wolfprovider-post-quantum-cryptography-ml-kem-and-ml-dsa/ - as of 2026-07-02
71. wolfSSL 5.9.2 release blog - wolfSSL - https://www.wolfssl.com/wolfssl-5-9-2-release-blog/ - as of 2026-06-30
72. wolfPKCS11 2.1.0 is out! (X post) - https://x.com/wolfSSL/status/2072742269816369452 - as of 2026-07-02
73. Java is now #PostQuantum ready with wolfSSL! (X post) - https://x.com/wolfSSL/status/2087299324380135712 - as of 2026-08-11

74. Post-Quantum Cryptography in curl - wolfSSL - https://www.wolfssl.com/post-quantum-cryptography-in-curl/ - as of 2026-06-16
75. Botan - https://botan.randombit.net/ - as of 2026-08-13
76. wolfssl.com — support for the official post quantum standards ml kem and ml dsa - https://www.wolfssl.com/support-for-the-official-post-quantum-standards-ml-kem-and-ml-dsa/
77. wolfssl.com — wolfcrypt post quantum - https://www.wolfssl.com/products/wolfcrypt-post-quantum/ - as of 2026-05-05
78. wolfssl.com — wolftpm add tpm 2 0 v1 85 pqc post quantum support - https://www.wolfssl.com/wolftpm-add-tpm-2-0-v1-85-pqc-post-quantum-support/ - as of 2026-01-15
79. wolfssl.com — the first firmware tpm with post quantum cryptography - https://www.wolfssl.com/the-first-firmware-tpm-with-post-quantum-cryptography/ - as of 2026-04-29
80. wolfssl.com — rust support for post quantum cryptography - https://www.wolfssl.com/rust-support-for-post-quantum-cryptography/ - as of 2026-03-12
81. bsi.bund.de — kryptobibliothek botan dvl - https://www.bsi.bund.de/EN/Themen/Unternehmen-und-Organisationen/Informationen-und-Empfehlungen/Kryptografie/Kryptobibliothek-Botan/kryptobibliothek-botan_dvl.html - as of 2025-06-02
82. botan.randombit.net — news - https://botan.randombit.net/news.html - as of 2025-02-04
83. botan.randombit.net — pubkey - https://botan.randombit.net/handbook/api_ref/pubkey.html
84. patchwork.ipfire.org — 20241109173814.1058040 4 adolf.belka@ipfire.org - https://patchwork.ipfire.org/project/ipfire/patch/20241109173814.1058040-4-adolf.belka@ipfire.org/
85. patchwork.yoctoproject.org — 20250217192141.1383722 1 peter.marko@siemens.com - https://patchwork.yoctoproject.org/project/oe-core/patch/20250217192141.1383722-1-peter.marko@siemens.com/ - as of 2025-02-07
86. gitlab.com — 1925 - https://gitlab.com/gnutls/gnutls/-/merge_requests/1925
87. docs.redhat.com — new features - https://docs.redhat.com/en/documentation/red_hat_enterprise_linux/9/html/9.8_release_notes/new-features
88. Blame · NEWS · master · gnutls / GnuTLS · GitLab - https://gitlab.com/gnutls/gnutls/-/blame/master/NEWS - as of 2026-04-29
89. en.wikipedia.org — GnuTLS - https://en.wikipedia.org/wiki/GnuTLS - as of 2026-04-29
90. firefox-source-docs.mozilla.org — nss 3 105 - https://firefox-source-docs.mozilla.org/security/nss/releases/nss_3_105.html - as of 2024-09-26
91. firefox-source-docs.mozilla.org — nss 3 118 - https://firefox-source-docs.mozilla.org/security/nss/releases/nss_3_118.html - as of 2025-11-18
92. en.wikipedia.org — Network Security Services - https://en.wikipedia.org/wiki/Network_Security_Services - as of 2026-05-15
93. arxiv.org — 2508.16078v1 - https://arxiv.org/html/2508.16078v1 - as of 2025-08-22

94. github.com — 1.0.22 RELEASE - https://github.com/jedisct1/libsodium/releases/tag/1.0.22-RELEASE - as of 2026-04-09
95. mbed-tls.readthedocs.io — roadmap - https://mbed-tls.readthedocs.io/en/latest/project/roadmap/
96. lists.trustedfirmware.org — D73QWEQBW5LHMFMG35S3IPV62NTKR4SO - https://lists.trustedfirmware.org/archives/list/mbed-tls@lists.trustedfirmware.org/thread/D73QWEQBW5LHMFMG35S3IPV62NTKR4SO/
97. oasis-open.org — oasis approves two public key cryptography standards - https://www.oasis-open.org/2026/07/15/oasis-approves-two-public-key-cryptography-standards/ - as of 2026-07-15
98. PQC Reference Library — Standards, Research Papers & Guides | PQC Today - https://www.pqctoday.com/library - as of 2026-06-03
99. docs.oasis-open.org — pkcs11 spec v3.2 - https://docs.oasis-open.org/pkcs11/pkcs11-spec/v3.2/pkcs11-spec-v3.2.html - as of 2026-06-03
100. ML-KEM Programming Guide - https://thalesdocs.com/gphsm/luna/7/docs/network/Content/sdk/extensions/pqc/ML-KEM_programming_guide.htm - as of 2026-07-17
101. Primus PKCS#11 API Provider - https://docs.securosys.com/pkcs/overview/ - as of 2026-08-04
102. PKCS#11 Library - https://support.fortanix.com/docs/fortanix-dsm-clients-pkcs11-library - as of 2026-07-24
103. Post-Quantum Cryptography in PKCS#11: ML-KEM and ML-DSA in wolfSSL and wolfPKCS11 - https://www.wolfssl.com/post-quantum-cryptography-in-pkcs11-ml-kem-and-ml-dsa-in-wolfssl-and-wolfpkcs11/ - as of 2026-07-10
104. Mastercard/pkcs11-tools - https://github.com/Mastercard/pkcs11-tools - as of 2026-08-22
105. github.com — pulls - https://github.com/open-quantum-safe/oqs-provider/pulls - as of 2026-08-04
106. cpl.thalesgroup.com — luna hsm pqc quantum safe encryption - https://cpl.thalesgroup.com/blog/encryption/luna-hsm-pqc-quantum-safe-encryption - as of 2025-07-29
107. data-protection-updates.gemalto.com — new luna pqc enhancements luna hsm firmware v7 9 1 and universal client v10 9 now available - https://data-protection-updates.gemalto.com/2025/09/24/new-luna-pqc-enhancements-luna-hsm-firmware-v7-9-1-and-universal-client-v10-9-now-available/ - as of 2025-09-24
108. thalestct.com — luna t series hardware security module 7 15 0 release announcement - https://www.thalestct.com/luna-t-series-hardware-security-module-7-15-0-release-announcement/ - as of 2025-09-17
109. cpl.thalesgroup.com — network hsms - https://cpl.thalesgroup.com/encryption/hardware-security-modules/network-hsms
110. github.com — SoftHSMv2 - https://github.com/softhsm/SoftHSMv2
111. github.com — 800 - https://github.com/softhsm/SoftHSMv2/issues/800 - as of 2025-07-22
112. dev.to — post quantum cryptography through pkcs11 in a software hsm 45if - https://dev.to/victorcraton/post-quantum-cryptography-through-pkcs11-in-a-software-hsm-45if - as of 2026-03-24

113. aws.amazon.com — aws kms post quantum ml dsa digital signatures - https://aws.amazon.com/about-aws/whats-new/2025/06/aws-kms-post-quantum-ml-dsa-digital-signatures - as of 2025-06-13
114. docs.keyfactor.com — post quantum cryptography keys and signatures - https://docs.keyfactor.com/ejbca/9.1.1/post-quantum-cryptography-keys-and-signatures
115. oasis-open.org — pkcs 11 specification version 3 2 - https://www.oasis-open.org/standard/pkcs-11-specification-version-3-2/ - as of 2025-11-14
116. docs.keyfactor.com — post quantum cryptography keys and signatures - https://docs.keyfactor.com/ejbca/latest/post-quantum-cryptography-keys-and-signatures
117. openssl-library.org — 2025 04 08 openssl 35 final release - https://openssl-library.org/post/2025-04-08-openssl-35-final-release/ - as of 2025-04-08
118. openquantumsafe.org — tls - https://openquantumsafe.org/applications/tls.html
119. github.com — oqs provider entrust - https://github.com/openca/oqs-provider-entrust
120. p11-glue.github.io — sharing - https://p11-glue.github.io/p11-glue/p11-kit/manual/sharing.html
121. Post-Quantum Cryptography Comes to Windows Insiders and Linux (Microsoft) - https://techcommunity.microsoft.com/blog/microsoft-security-blog/post-quantum-cryptography-comes-to-windows-insiders-and-linux/4413803 - as of 2025-05-21
122. directaccess.richardhicks.com — microsoft ad cs adds post quantum cryptography support with ml dsa - https://directaccess.richardhicks.com/2026/05/18/microsoft-ad-cs-adds-post-quantum-cryptography-support-with-ml-dsa/ - as of 2026-05-18
123. encryptionconsulting.com — ml dsa support for your microsoft pki - https://www.encryptionconsulting.com/ml-dsa-support-for-your-microsoft-pki/ - as of 2026-06-30
124. windowsforum.com — kb5101650 adds windows 11 hybrid ml kem tls 1 3 key exchange.438638 - https://windowsforum.com/windows-news.4/kb5101650-adds-windows-11-hybrid-ml-kem-tls-1-3-key-exchange.438638/ - as of 2026-07-14
125. paragmali.com — cng architecture bcrypt ncrypt ksps - https://paragmali.com/blog/cng-architecture-bcrypt-ncrypt-ksps/ - as of 2026-05-16
126. docs.oracle.com — oracle providers - https://docs.oracle.com/en/java/javase/24/security/oracle-providers.html
127. seanjmullan.org — jdk24 - https://seanjmullan.org/blog/2025/04/07/jdk24 - as of 2025-04-07
128. seanjmullan.org — jdk25 - https://seanjmullan.org/blog/2025/09/23/jdk25 - as of 2025-09-23
129. Exploiting ML-DSA bugs (D. J. Bernstein) - https://cr.yp.to/papers/mldsa-20260601.pdf - as of 2026-06-04
130. Breaking Optimized HQC: The First Cache-Timing Full Decryption Oracle Key-Recovery… (Crypto 2026) - https://www.youtube.com/watch?v=8_FGGAHKgdI - as of 2026-08-15
131. OpenSSL Security Advisory [25th August 2026] - https://openssl-library.org/news/secadv/20260825.txt - as of 2026-08-25

132. liboqs/RELEASE.md at main · open-quantum-safe/liboqs - https://github.com/open-quantum-safe/liboqs/blob/main/RELEASE.md - as of 2026-07-09
133. A Survey of Post-Quantum Cryptography Support in Cryptographic Libraries - https://arxiv.org/html/2508.16078v1 - as of 2026-08-15
134. Google Boringssl : Security vulnerabilities, CVEs - https://www.cvedetails.com/vulnerability-list/vendor_id-1224/product_id-47071/Google-Boringssl.html - as of 2026-08-15
135. Latest NIST PQC Standards and more - Bouncy Castle Java 1.79 - Bouncycastle - https://www.bouncycastle.org/resources/latest-nist-pqc-standards-and-more-bouncy-castle-java-1-79/ - as of 2026-08-15
136. CVE-2025-52473 | Tenable® - https://www.tenable.com/cve/CVE-2025-52473 - as of 2025-07-10
137. Releases · open-quantum-safe/liboqs - https://github.com/open-quantum-safe/liboqs/releases - as of 2025-07-10
138. PQCA announces release of liboqs version 0.14.0 from Open Quantum Safe project – Post-Quantum Cryptography Alliance - https://pqca.org/blog/2025/pqca-announces-release-of-liboqs-version-0-14-0-from-open-quantum-safe-project/ - as of 2025-07-10
139. Theoretical design flaw in HQC · Advisory · open-quantum-safe/liboqs · GitHub - https://github.com/open-quantum-safe/liboqs/security/advisories/GHSA-3rxw-4v8q-9gq5
140. Securityvulnerability - https://securityvulnerability.io/vulnerability/CVE-2025-48946
141. Using AISLE to Find Vulnerabilities in Amazon's Crypto Stack: AWS-LC and s2n-tls - https://aisle.com/blog/aisle-finds-vulnerabilities-in-amazons-crypto-stack-aws-lc-and-s2n-tls - as of 2026-03-03
142. Amazon AWS-LC Vulnerability Allows Attackers to Bypass Certificate Chain Verification - https://cyberpress.org/amazon-aws-lc-vulnerability/ - as of 2026-03-06
143. aws-lc/crypto/fipsmodule/PQREADME.md at main · aws/aws-lc - https://github.com/aws/aws-lc/blob/main/crypto/fipsmodule/PQREADME.md
144. AWS Security Blog - https://aws.amazon.com/blogs/security/aws-lc-fips-3-0-first-cryptographic-library-to-include-ml-kem-in-fips-140-3-validation
145. BoringSSL through 2018-06-14 allows a memory-cache side... · CVE-2018-12440 · GitHub Advisory Database · GitHub - https://github.com/advisories/GHSA-mwrj-3qqh-xprh - as of 2018-06-14
146. NIST PQC Support and more - Bouncy Castle C# .NET 2.5.0 - Bouncycastle - https://www.bouncycastle.org/resources/nist-pqc-support-and-more-bouncy-castle-c-net-2-5-0/ - as of 2024-12-06
147. Bouncy Castle Releases Java 1.81 and C# .NET 2.6.1 - Bouncycastle - https://www.bouncycastle.org/resources/bouncy-castle-releases-java-1-81-and-c-net-2-6-1/ - as of 2025-06-09
148. New Releases: Bouncy Castle Java 1.84 and Bouncy Castle Java LTS 2.73.11 - https://www.bouncycastle.org/resources/new-releases-bouncy-castle-java-1-84-and-bouncy-castle-java-lts-2-73-11/ - as of 2026-05-12
149. Vulnerabilities | OpenSSL Library - https://openssl-library.org/news/vulnerabilities/ - as of 2026-04-07

150. OpenSSL 3.5 Series Release Notes | OpenSSL Library - https://openssl-library.org/news/openssl-3.5-notes/ - as of 2025-04-08

151. OpenSSL 3.5 Released » ADMIN Magazine - https://www.admin-magazine.com/News/OpenSSL-3.5-Released

152. OpenSSL Position and Plans on Private Key Formats for the ML-KEM and ML-DSA Post-quantum (PQ) Algorithms | OpenSSL Corporation - https://openssl-corporation.org/post/2025-01-21-blog-positionandplans/ - as of 2025-01-21